BIOPULSE USER STORIES

User Stories Overview

Purpose

User stories were developed from the personas identified during the requirements analysis phase of BioPulse. These stories translate user needs into functional system requirements and provide a bridge between user expectations and interface design.

The user stories guided the development of emergency workflows, monitoring features, accessibility requirements, and high-fidelity interface designs implemented throughout the BioPulse prototype.

The user stories were primarily derived from the three personas:

- Sarah Williams (Vulnerable Commuter)
- Michael Chen (Independent Traveller)
- Margaret Taylor (Independent Senior)

USER STORY 1: Create an Account

Related Persona(s)

- Sarah Williams
- Margaret Taylor

User Story

As a first-time user, I want to create an account so that my personal information and emergency settings can be stored securely.

Acceptance Criteria

- User can enter their full name
- User can enter their email address
- User can create a password
- User receives confirmation of successful account creation

Related Screens

- Welcome Screen
- Create Account Screen

Design Rationale

A simple registration process reduces onboarding friction and allows users to quickly gain access to emergency monitoring features.

USER STORY 2: Secure User Login

Related Persona(s)

- Sarah Williams
- Michael Chen
- Margaret Taylor

User Story

As a returning user, I want to securely log into my account so that I can access BioPulse monitoring features and emergency functions.

Acceptance Criteria

- User enters email address
- User enters password
- Authentication succeeds
- Monitoring dashboard becomes accessible

Related Screens

- Welcome Screen
- Login Screen

Design Rationale

Using familiar login patterns improves learnability and allows users to quickly access the application.

USER STORY 3: Password Recovery

Related Persona(s)

- Sarah Williams
- Margaret Taylor

User Story

As a user, I want to reset my password if I forget it so that I can regain access to my account.

Acceptance Criteria

- User enters registered email address
- Password reset request is submitted
- User can return to the Login Screen

Related Screens

- Login Screen
- Forgot Password Screen

Design Rationale

Providing a recovery process prevents account lockout and improves overall usability.

USER STORY 4: Pair Wearable Device

Related Persona(s)

- Sarah Williams
- Michael Chen

User Story

As a user, I want to pair my SkinGrip wearable quickly so that physiological monitoring can begin with minimal setup.

Acceptance Criteria

- Wearable detected using NFC
- Successful pairing confirmation displayed
- Device connection status visible
- User proceeds to monitoring dashboard

Related Screens

- NFC Pairing Screen

Design Rationale

Fast NFC pairing reduces setup complexity and encourages continued use of the monitoring system.

USER STORY 5: Continuous Physiological Monitoring

Related Persona(s)

- Sarah Williams
- Michael Chen

User Story

As a user, I want my physiological signals continuously monitored so that abnormal activity can be identified automatically.

Acceptance Criteria

- GSR readings displayed
- Movement readings displayed
- Monitoring status visible
- Background monitoring remains active

Related Screens

- Normal Monitoring Dashboard

Design Rationale

Continuous monitoring provides users with reassurance while enabling automatic emergency detection.

USER STORY 6: Threat Detection Notification

Related Persona(s)

- Sarah Williams
- Michael Chen

User Story

As a user, I want BioPulse to notify me when unusual physiological activity is detected so that I can respond appropriately.

Acceptance Criteria

- GSR abnormality detected
- Threat notification displayed
- Emergency countdown initiated
- Threat status clearly visible

Related Screens

- GSR Spike Detection Screen

Design Rationale

Early warning allows users to review potential threats before emergency escalation occurs.

USER STORY 7: Cancel a False Alarm

Related Persona(s)

- Sarah Williams
- Michael Chen

User Story

As a user, I want to cancel accidental emergency alerts using Face ID so that emergency services are not contacted unnecessarily.

Acceptance Criteria

- Face ID authentication available
- Alert can be cancelled
- Monitoring continues normally
- Cancellation confirmation displayed

Related Screens

- Threat Detection Countdown Screen
- Threat Cancellation Screen

Design Rationale

Biometric verification reduces accidental cancellations while preventing unnecessary emergency responses.

USER STORY 8: Emergency Escalation

Related Persona(s)

- Sarah Williams
- Margaret Taylor

User Story

As a vulnerable traveller, I want BioPulse to automatically contact emergency services and share my location so that help can arrive quickly if I am unable to respond.

Acceptance Criteria

- SOS alert transmitted
- GPS coordinates attached
- Emergency contacts notified
- Dispatch confirmation displayed

Related Screens

- Threat Detection Countdown Screen
- Active Emergency Screen

Design Rationale

Automatic escalation ensures emergency assistance can be requested even when the user cannot manually operate their device.

USER STORY TRACEABILITY MATRIX

User Story	Persona	Related Screen
Create Account	Sarah, Margaret	Welcome Screen, Create Account Screen
Login	All Personas	Welcome Screen, Login Screen
Password Recovery	Sarah, Margaret	Login Screen, Forgot Password Screen
Pair Wearable	Sarah, Michael	NFC Pairing Screen
Physiological Monitoring	Sarah, Michael	Monitoring Dashboard
Threat Detection	Sarah, Michael	GSR Spike Detection Screen
False Alarm Cancellation	Sarah, Michael	Countdown Screen, Cancellation Screen
Emergency Escalation	Sarah, Margaret	Countdown Screen, Active Emergency Screen

Persona-to-Screen Mapping

Sarah Williams (Vulnerable Commuter)

Screens Influenced:

- Login Screen
- Monitoring Dashboard
- Threat Detection Screen
- Countdown Screen
- Emergency Screen

Michael Chen (Independent Traveller)

Screens Influenced:

- NFC Pairing Screen
- Monitoring Dashboard
- Threat Detection Screen
- Countdown Screen

Margaret Taylor (Independent Senior)

Screens Influenced:

- Welcome Screen
- Create Account Screen
- Login Screen
- Emergency Screen

Conclusion

The BioPulse user stories translate persona requirements into tangible system functionality. These stories directly informed screen design, navigation flows, emergency workflows, accessibility requirements, and the overall user experience of the BioPulse prototype. Together, they establish traceability between user needs and implemented interface features.